Percentage Yield & Purity

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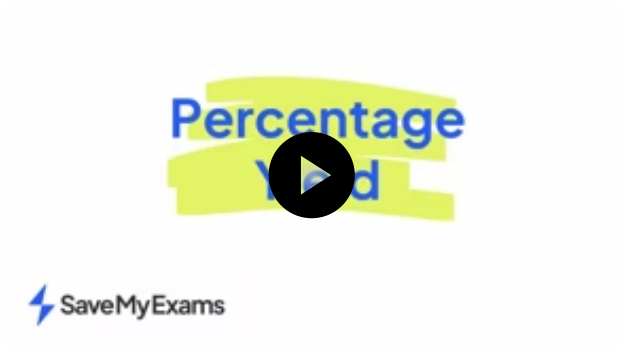

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Calculating percentage yield, percentage by mass & percentage purity

Extended tier only

Percentage yield

- **Yield** is the term used to describe the amount of **product** you get from a reaction
- For economic reasons, the objective of every chemical producing company is to have as high a percentage yield as possible to increase profits and reduce costs and waste
- In practice, you **never** get 100% yield in a chemical process for several reasons
- These include:
  - Some reactants may be left behind in the **equipment**
  - The reaction may be **reversible** and in these reactions a high yield is never possible as the products are continually turning back into the reactants
  - Some products may also be lost during **separation** and **purification** stages such as filtration or distillation
  - There may be **side reactions** occurring where a substance reacts with a gas in the air or an **impurity** in one of the reactants

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

How to calculate percentage yield

- **Percentage yield** compares the actual yield to the theoretical yield
- The equation for percentage yield is:

$$\text{percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

- **Actual yield** is the recorded amount of product obtained
- **Theoretical yield** is the amount of product that would be obtained under perfect practical and chemical conditions
  - Typically, this involves a reacting mass calculation based on the balanced symbol equation

Worked Example

Copper(II) sulfate may be prepared by the reaction of dilute sulfuric acid with copper(II) oxide. A student prepared 1.6 g of dry copper(II) sulfate crystals.

Calculate the percentage yield if the theoretical yield is 2.0 g.

Answer

- Actual yield of copper(II) sulfate = 1.6 g
- Percentage yield of copper(II) sulfate = $\frac{1.6}{2.0}$ x 100 = **80%**

Examiner Tips and Tricks

- You are expected to remember the equation for percentage yield
- If you remember it incorrectly and get a percentage yield greater than 100%, then you have made an error!
- The most common error is to divide the theoretical yield by the actual yield
  - In this case, you just need to swap the numbers around in your calculation

How to calculate percentage composition by mass

- The percentage composition of any compound is a way to express the mass of each element as a percentage of the total mass of the compound
- The equation for percentage composition is:

$$\text{percentage composition of an element} = \frac{\text{total mass of the element in the compound}}{\text{relative formula mass of the compound}} \times 100$$

- For example, in water:

oxygen (O) atom
    - From the Periodic Table, the relative atomic mass of:
    - Hydrogen = 1
    - Oxygen = 16
- Therefore, the total mass of water is:
    - $(2 \times 1) + 16 = 18$
- To find the percentage composition of hydrogen:
    - Percentage of hydrogen = $\frac{2 \times 1}{18} \times 100 = 11.1\%$
- Similarly, the percentage composition of oxygen is:
    - Percentage of oxygen = $\frac{1 \times 16}{18} \times 100 = 88.9\%$
- **Note:** The total percentage by mass of all the elements should add up to 100%, e.g. 11.1% + 88.9% = 100%

Worked Example

Calculate the percentage by mass of iron in iron(III) oxide, Fe_2O_3.

Answer:

- From the Periodic Table, the relative atomic masses are:
    - Fe = 56
    - Oxygen = 16
- The total mass of iron in iron(III) oxide is:
    - 2 x 56 = 112
- The total mass of iron(III) oxide is:
    - $(2 \times 56) + (3 \times 16) = 160$
- The equation for percentage composition is:
    - Percentage composition = $\frac{\text{total mass of element}}{\text{total mass of compound}} \times 100$
- So, the percentage composition of iron in iron(III) oxide is:
    - Percentage of iron = $\frac{112}{160} \times 100 = 70\%$

Worked Example

The chemical formula of the fertiliser ammonium nitrate is NH_4NO_3. Calculate the percentage by mass of nitrogen in ammonium nitrate.

Answer:

- From the Periodic Table, the relative atomic masses are:
    - Nitrogen = 14
    - Hydrogen = 1
    - Oxygen = 16
- **Careful:** There are **two** nitrogen atoms in ammonium nitrate
- The total mass of nitrogen in ammonium nitrate is:
    - 2 x 14 = 28

- Percentage Composition = $\frac{\text{total mass of element}}{\text{total mass of compound}} \times 100$
- So, the percentage composition of nitrogen in ammonium nitrate is:
- Percentage of nitrogen = $\frac{(2 \times 14)}{80} \times 100 = 35\%$

Examiner Tips and Tricks

- Make sure you calculate the percentage composition using the **total mass** of the element.
  - For example, a common mistake with ammonium nitrate is doing the calculation for only one atom of nitrogen.
  - This would lose a mark in an exam
- Show ALL your working out. If you make a mistake in the calculation, you could still score a mark for your workings.

How to calculate percentage purity

- A pure substance has nothing else mixed with it
- Often, the product you are trying to obtain may become contaminated with unwanted substances such as unreacted reactants, catalysts and other impurities
- To equation to calculate percentage purity is:

$$\text{percentage purity} = \frac{\text{mass of pure substance}}{\text{total mass of substance}} \times 100$$

Worked Example

A sample of lead(II) bromide was made. It weighed 15 g.

The sample was found to be impure and only contained 13.5 g of lead(II) bromide.

Calculate the percentage purity of the lead(II) bromide.

Answer:

- The mass of the pure substance is 13.5 g
- The total mass of the substance is 15 g
- Percentage purity = $\frac{\text{mass of pure substance}}{\text{total mass of substance}} \times 100$
- Percentage purity = $\frac{13.5}{15} \times 100 = \mathbf{90\%}$

Examiner Tips and Tricks

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